

# TBD

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**Abstract—Abstract**

**Index Terms—Keywords**

## I. INTRODUCTION

Reference [1]

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## II. BACKGROUND

### III. MODEL

This section presents the simulation models for comparing proactive (BGP-like) and reactive (DNS-like) approaches for distributing Endpoint Identifier (EID) reachability information in Delay-Tolerant Networks (DTNs). Both protocols are implemented in OMNeT++ [2] and operate over identical physical topologies to ensure a fair comparison.

#### A. System Model

We consider a DTN consisting of  $N$  nodes interconnected by links with propagation delay  $d_{ij}$  between nodes  $i$  and  $j$ . Each node may host Bundle Protocol (BP) agents identified by unique EIDs [3]. The fundamental problem is: *how should nodes discover the Convergence-Layer Adapter (CLA) endpoint responsible for a given EID?*

Let  $\mathcal{E} = \{e_1, e_2, \dots, e_M\}$  denote the set of EIDs in the network. For each EID  $e_k$ , there exists exactly one authoritative endpoint binding:

$$\text{endpoint}(e_k) = \langle \text{nodeId}, \text{claProtocol}, \text{port} \rangle \quad (1)$$

The goal is to enable any node  $i$  to resolve  $\text{endpoint}(e_k)$  with:

- **Correctness:** The resolved endpoint matches the current authoritative binding
- **Timeliness:** Resolution completes within acceptable latency bounds
- **Efficiency:** Minimal network overhead and state distribution

Nodes in the network can assume different roles:

- **Publishers:** Nodes that originate EID bindings and announce them to the network
- **Clients:** Nodes that need to resolve EIDs to their endpoint bindings
- **Resolvers** (DNS only): Caching intermediaries that handle client queries
- **Authorities** (DNS only): Nodes that store authoritative EID records

#### B. Network Model

The primary experimental topology is a two-dimensional grid network:

$$G = (V, E) \text{ where } |V| = w \times h \quad (2)$$

Nodes are indexed  $v_{i,j}$  for  $i \in [0, w-1], j \in [0, h-1]$ , with edges connecting adjacent nodes:

$$E = \{(v_{i,j}, v_{i',j'}) : |i - i'| + |j - j'| = 1\} \quad (3)$$

The network diameter (maximum shortest path) for a  $w \times h$  grid is  $D = w + h - 2$ . For a  $10 \times 10$  grid,  $D = 18$  hops.

All links are modeled as delay channels with configurable propagation delay  $d$ . The default delay is  $d = 10$  ms for terrestrial links, while deep-space scenarios use  $d \in \{1, 5, 10, 20\}$  s. The channel model abstracts transmission time, focusing on propagation delay as appropriate for control-plane traffic with small message sizes.

For realistic scenarios, we model hierarchical topologies representing terrestrial disaster response (5–30 ms delays), lunar exploration networks (5 ms to 1.3 s delays), and Mars exploration networks (1 s to 720 s delays). These topologies include specialized channel types such as Earth-Moon links (1.3 s one-way light delay), Mars orbital relay channels (60 s), and interplanetary channels (720 s average).

#### C. BGP-like Push Model

The BGP-like model implements a path-vector routing protocol inspired by BGP-4 [4]. EID bindings are proactively flooded to all nodes upon publication, ensuring that each node maintains a local routing table with all known EIDs.

1) *Data Structures:* Each node maintains a routing table indexed by EID:

$$\text{bgpTable} : \mathbb{Z}^+ \rightarrow \text{BgpEntry} \quad (4)$$

where each entry contains:

- **eid:** The Endpoint Identifier
- **originId:** Publishing node ID
- **nextHop:** Gate index for forwarding
- **pathLength:** Distance in hops from the origin
- **version:** Update sequence number for churn handling
- **endpoint:** CLA binding information
- **path:** Full path vector for loop detection

2) *Message Types:* The protocol defines two message types:

- **BgpAnnounce** (64 + 4|path| bytes): Propagates EID reachability with path vector
- **BgpWithdraw** (32 bytes): Removes EID reachability

3) *Publication Algorithm*: When node  $p$  publishes EID  $e$  with endpoint  $\epsilon$ :

- 1) Create local table entry with  $\text{pathLength} = 0$  and  $\text{path} = [p]$
- 2) Increment version:  $v \leftarrow \text{publishedVersions}[e] + 1$
- 3) Construct announcement:  $m \leftarrow \text{BgpAnnounce}(e, p, 0, v, \epsilon, t_{\text{now}}, [p])$
- 4) Flood to all neighbors:  $\forall g \in \text{neighborGates} : \text{send}(m, g)$
- 4) *Announcement Processing*: Upon receiving announcement  $m$  at node  $n$  from gate  $g_{\text{in}}$ :

**Step 1: Loop Detection.** If  $n \in m.\text{path}$ , discard the message.

**Step 2: Path Extension.** Compute  $\text{path}' \leftarrow m.\text{path} \cup \{n\}$  and  $\text{pathLength}' \leftarrow m.\text{pathLength} + 1$ .

**Step 3: Acceptance Decision.** Let  $\text{existing} = \text{bgpTable}[m.\text{eid}]$ . Accept if any condition holds:

$$\text{accept}(m) = \begin{cases} \text{true} & \text{if } \nexists \text{existing} \\ \text{true} & \text{if } m.\text{originId} = \text{existing}.\text{originId} \\ & \quad \wedge m.\text{version} > \text{existing}.\text{version} \\ \text{true} & \text{if } \text{pathLength}' < \text{existing}.\text{pathLength} \\ \text{true} & \text{if } \text{pathLength}' = \text{existing}.\text{pathLength} \\ & \quad \wedge m.\text{originId} < \text{existing}.\text{originId} \\ \text{false} & \text{otherwise} \end{cases} \quad (5)$$

**Step 4: Table Update and Forwarding.** If accepted, update the local table and forward to all neighbors except the arrival gate.

5) *Withdrawal Processing*: Withdrawal  $w$  for EID  $e$  is processed only if  $w.\text{originId}$  matches the current entry's origin and  $w.\text{version} \geq \text{existing}.\text{version}$ . Upon acceptance, the entry is removed and the withdrawal is forwarded.

6) *Complexity Analysis*: **Convergence Time**: For a network with diameter  $D$  and link delay  $d$ :

$$T_{\text{conv}}^{\text{BGP}} = D \cdot d \quad (6)$$

For a  $10 \times 10$  grid with  $d = 10$  ms:  $T_{\text{conv}}^{\text{BGP}} = 18 \times 0.01 = 0.18$  s.

**State Complexity**: Per-node state is  $O(M)$  where  $M$  is the number of EIDs. Network-wide state is  $O(N \cdot M)$ —full replication.

**Message Complexity**:  $O(N)$  messages per EID publication in a grid topology.

#### D. DNS-like Pull Model

The DNS-like model implements an on-demand resolution protocol inspired by the Domain Name System [5]. EID bindings are stored at authoritative servers and resolved via queries with optional caching.

1) *Data Structures*: **Authority Record** (at authority nodes):

- **eid**: The Endpoint Identifier
- **publisherId**: Publisher node ID
- **endpoint**: CLA binding information
- **ttl**: Time-to-live in seconds
- **version**: Update sequence number
- **lastChangedTime**: Timestamp of last modification

**Cache Entry** (at resolver nodes):

- **eid**: The Endpoint Identifier
- **endpoint**: Cached CLA binding
- **ttl**: Original TTL value
- **expiryTime**: When the cache entry expires
- **recordTime**: Authority's lastChangedTime (for staleness detection)

2) *Message Types*:

- **DnsQuery** (48 bytes): Client query for an EID
- **DnsResponse** (80 bytes): Response with endpoint binding
- **DnsRegister** (72 bytes): Publisher registration at authority
- **DnsDeregister** (24 bytes): Publisher withdrawal

3) *Registration*: When publisher  $p$  registers EID  $e$  at authority  $a$ :

If  $p = a$  (local authority), the record is stored directly. Otherwise, a **DnsRegister** message is sent to the authority. Authority assignment uses modular hashing:

$$a = (e \bmod |\mathcal{A}|) + a_{\text{first}} \quad (7)$$

where  $\mathcal{A}$  is the set of authority nodes.

4) *Query Resolution*: Client-initiated query for EID  $e$ :

- 1) **Local Cache Check**: If  $e \in \text{dnsCache}$  and not expired, return cache hit
- 2) **Cache Miss**: Create query and forward to resolver
- 3) **Resolver Processing**: Check resolver cache; if miss, forward to authority
- 4) **Authority Processing**: Look up record and return response
- 5) **Response Path**: Response traverses back, with resolvers caching the result

5) *Caching Mechanism*: Cache entries are inserted with expiry time  $t_{\text{now}} + \tau$  where  $\tau$  is the TTL. When the cache exceeds capacity, expired entries are removed first, then the entry with earliest expiry time is evicted.

The cache hit rate is:

$$\text{hitRate} = \frac{\sum \text{cacheHits}}{\sum \text{cacheHits} + \sum \text{cacheMisses}} \quad (8)$$

6) *Complexity Analysis*: **Query Latency (cache miss)**:

$$L_{\text{query}} \approx 2 \cdot D \cdot d \quad (9)$$

**State Complexity**: Authority state is  $O(M)$ . Resolver cache is  $O(\min(M, C))$  where  $C$  is cache capacity. Network-wide state is  $O(M)$ —centralized.

**Message Complexity**:  $O(1)$  messages per registration;  $O(1)$  per query.

#### E. Fair Comparison Design

To ensure a fair comparison, both protocols operate over the same physical grid topology. However, DNS in practice runs over an already-routed IP network, whereas BGP-like flooding operates at the network layer. To account for this, we implement a “direct mode” for DNS.

1) *Direct Mode*: In direct mode (`dnsDirectMode=true`), DNS queries use OMNeT++'s `sendDirect()` mechanism with RTT-based delays:

$$\text{RTT}_{\text{DNS}} = 2 \cdot D \cdot d \quad (10)$$

This models DNS running over a pre-routed IP underlay, isolating the protocol comparison from routing convergence effects. The approach ensures that:

- BGP announcements propagate through the physical topology hop-by-hop
- DNS queries logically traverse: Client  $\rightarrow$  Resolver  $\rightarrow$  Authority  $\rightarrow$  Resolver  $\rightarrow$  Client
- Both protocols experience comparable network latencies

#### F. Staleness and Accuracy Model

A central `GroundTruth` module maintains authoritative records for all EIDs, enabling correctness verification.

1) *BGP Accuracy*: A node's BGP entry for EID  $e$  is *correct* if:

$$\text{correct}_{\text{BGP}}(n, e) = \begin{cases} \text{bgpTable}[e].\text{endpoint} = \text{GT}[e].\text{endpoint} & \text{if active} \\ e \notin \text{bgpTable} & \text{if withdrawn} \end{cases} \quad (11)$$

Network-wide BGP accuracy:

$$\text{Accuracy}_{\text{BGP}}(e) = \frac{|\{n : \text{correct}_{\text{BGP}}(n, e)\}|}{N} \quad (12)$$

2) *DNS Staleness*: A DNS response is *stale* if the cached record predates the last authoritative change:

$$\text{stale}(r) = r.\text{recordTime} < \text{GT}[r.\text{eid}].\text{lastChangeTime} \quad (13)$$

Under churn with interval  $\Delta_{\text{churn}}$ :

$$P(\text{stale}) \approx \frac{\Delta_{\text{churn}}}{\tau} \text{ for } \tau > \Delta_{\text{churn}} \quad (14)$$

where  $\tau$  is the TTL.

3) *Convergence Detection*: Convergence for EID  $e$  occurs when  $\geq 99\%$  of nodes have correct information:

$$\text{converged}(e, t) = \frac{|\{n : \text{correct}(n, e, t)\}|}{N} \geq 0.99 \quad (15)$$

We distinguish *initial convergence* (first publication) from *churn convergence* (subsequent updates).

#### G. Query Distribution Model

Query patterns follow Zipf's law [6]:

$$P(\text{query for EID } e_k) \propto \frac{1}{k^\alpha} \quad (16)$$

where  $k$  is the rank (1 = most popular) and  $\alpha$  is the skewness parameter. Values of  $\alpha = 0$  yield uniform distribution, while  $\alpha = 1.0$  approximates typical web traffic patterns.

#### H. Metrics Summary

Table I summarizes the key metrics collected during simulation.

TABLE I  
SIMULATION METRICS

| Category | Metric            | Unit    |
|----------|-------------------|---------|
| Overhead | Total bytes sent  | bytes   |
|          | Total messages    | count   |
| Latency  | Discovery latency | seconds |
|          | Convergence time  | seconds |
|          | Query latency     | seconds |
| Accuracy | BGP accuracy      | [0, 1]  |
|          | DNS accuracy      | [0, 1]  |
|          | Stale rate        | [0, 1]  |
| State    | BGP table size    | entries |
|          | DNS cache size    | entries |

TABLE II  
PROTOCOL COMPLEXITY COMPARISON

| Aspect             | BGP (Push)          | DNS (Pull)                     |
|--------------------|---------------------|--------------------------------|
| State per node     | $O(M)$              | $O(1)$ client, $O(C)$ resolver |
| Network state      | $O(N \cdot M)$      | $O(M)$                         |
| Publication cost   | $O(N)$ messages     | $O(1)$ messages                |
| Query cost         | $O(1)$ (local)      | $O(D)$ hops                    |
| Churn cost         | $O(N)$ per change   | $O(1)$ per change              |
| Convergence time   | $O(D \cdot d)$      | N/A (on-demand)                |
| Discovery latency  | $D \cdot d$         | $2 \cdot D \cdot d$            |
| Resolution latency | $\approx 0$ (local) | $2 \cdot D \cdot d$            |

#### I. Complexity Comparison

Table II compares the theoretical complexity of both approaches.

## IV. EVALUATION

This section presents the evaluation methodology and experimental scenarios used to compare BGP-like and DNS-like protocols across different network environments.

#### A. Realistic Deployment Scenarios

Beyond synthetic grid topologies, we evaluate three realistic DTN deployment scenarios spanning the latency spectrum from terrestrial to interplanetary networks. Each scenario represents a plausible use case with hierarchical topology and heterogeneous link delays.

1) *Experiment 11: Terrestrial Disaster Response*: This scenario models a disaster response network with mobile command centers, relay vehicles, field teams, and reconnaissance drones. The network exhibits low latency (5–30 ms) and moderate churn as teams move through the affected area.

**Topology**: 22 nodes in a 4-tier hierarchy:

- Tier 1: Command Center (central authority/resolver)
- Tier 2: 3 Mobile Relay Vehicles
- Tier 3: 6 Field Teams (publishers)
- Tier 4: 12 Drones/Sensors (publishers)

Table III shows the DNS query round-trip times from each node type to the authority at the Command Center.

**Expected outcome**: DNS should outperform BGP due to low query latency ( $< 100$  ms RTT) and reduced overhead under churn.

TABLE III  
TERRESTRIAL NETWORK: DNS RTT TO AUTHORITY

| Node Type | Path to Command             | RTT   |
|-----------|-----------------------------|-------|
| Command   | Local                       | 0 ms  |
| Relay     | 10 ms $\times$ 2            | 20 ms |
| Team      | (20 + 10) ms $\times$ 2     | 60 ms |
| Drone     | (5 + 20 + 10) ms $\times$ 2 | 70 ms |

TABLE IV  
LUNAR NETWORK: DNS RTT TO GATEWAY AUTHORITY

| Node Type    | Path to Gateway              | RTT    |
|--------------|------------------------------|--------|
| Gateway      | Local                        | 0 s    |
| Earth DSN    | 1.3 s $\times$ 2             | 2.6 s  |
| Surface Base | 250 ms $\times$ 2            | 0.5 s  |
| Rover        | (250 + 10) ms $\times$ 2     | 0.52 s |
| Field Asset  | (250 + 10 + 5) ms $\times$ 2 | 0.53 s |

2) *Experiment 12: Lunar Artemis Network:* This scenario models a lunar exploration network supporting the Artemis program, connecting Earth ground stations, the Lunar Gateway orbital relay, and surface assets. The key challenge is the 1.3 s Earth-Moon light delay combined with near-instantaneous surface communications.

**Topology:** 12 nodes spanning Earth and Moon:

- Tier 0: Earth DSN (Earth authority)
- Tier 1: Lunar Gateway (lunar authority/resolver)
- Tier 2: 2 Surface Bases (South Pole, Equatorial)
- Tier 3: 4 Rovers
- Tier 4: 4 EVA suits/Drones

Table IV shows the DNS query RTT to the Lunar Gateway authority.

**Expected outcome:** This is the crossover point. For lunar-local queries ( $\sim 0.5$  s RTT), DNS may be acceptable. For Earth-involved queries ( $> 2.6$  s RTT), BGP's proactive model becomes attractive.

3) *Experiment 13: Mars Exploration Network:* This scenario models a Mars exploration network with 12-minute average Earth-Mars delay and significant surface relay delays. Per-query DNS latency becomes prohibitive.

**Topology:** 10 nodes spanning Earth and Mars:

- Tier 0: Earth DSN (Earth authority)
- Tier 1: Mars Relay Orbiter (Mars authority/resolver)
- Tier 2: 2 Surface Habitats (Jezero, Olympus)
- Tier 3: 4 Rovers
- Tier 4: 2 Deployed Instruments

Table V shows the DNS query RTT to the Mars Orbiter authority.

**Expected outcome:** BGP should decisively outperform DNS. Even Mars-local DNS queries require  $\sim 2$  minutes RTT, while BGP convergence pays once and enables instant local lookups thereafter.

### B. Summary of Realistic Scenarios

Table VI summarizes the three realistic scenarios and their key characteristics.

TABLE V  
MARS NETWORK: DNS RTT TO ORBITER AUTHORITY

| Node Type    | Path to Orbiter           | RTT             |
|--------------|---------------------------|-----------------|
| Mars Orbiter | Local                     | 0 s             |
| Earth DSN    | 720 s $\times$ 2          | 1440 s (24 min) |
| Habitat      | 60 s $\times$ 2           | 120 s           |
| Rover        | (60 + 5) s $\times$ 2     | 130 s           |
| Instrument   | (60 + 5 + 1) s $\times$ 2 | 132 s           |

TABLE VI  
REALISTIC SCENARIO SUMMARY

| Parameter       | Terrestrial | Lunar      | Mars    |
|-----------------|-------------|------------|---------|
| Nodes           | 22          | 12         | 10      |
| Link delays     | 5–30 ms     | 5 ms–1.3 s | 1–720 s |
| Max DNS RTT     | 70 ms       | 2.6 s      | 24 min  |
| Churn interval  | 30–60 s     | 120 s      | 300 s   |
| DNS TTL         | 30 s        | 60 s       | 120 s   |
| Expected winner | DNS         | Contested  | BGP     |

## V. CONCLUSIONS

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